A

REPORT

ON

"Android College News and Announcement Application"

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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This is to certify that report entitled

"Android College News and Announcement Application"

Is submitted as partial fulfillment of curriculum of the S.E. of E&C

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"Android College News and

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Place: Yeola **Prof.S.R.Aher**

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Abstract

The **College News Hub App** is an Android application developed using Java and Firebase to provide a digital platform for sharing college news and announcements. It replaces traditional notice boards with real-time updates accessible through smartphones. The system supports two user roles: Admin and Student, with secure login using Firebase Authentication. Admins can post, edit, and delete notices, while students can view, search, and bookmark updates. Firebase Firestore and Storage are used for data management, and Firebase Cloud Messaging enables push notifications. The application ensures fast, secure, and organized communication within the college.

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List of Abbreviations

Abbreviation	Description
UI	User Interface
DB	Database
API	Application Programming Interface
FCM	Firebase Cloud Messaging
OOP	Object Oriented Programming
XML	Extensible Markup Language
SDK	Software Development Kit

1. Introduction

College News Hub App is an Android-based application developed using **Java** and **Firebase** as backend. The application is designed to digitally manage and distribute college news, announcements, notices, event updates, exam schedules, and important alerts for students and staff.

Traditional notice boards are slow, manual, and inefficient. Important updates can be missed easily. This system solves that issue by providing instant digital notifications directly to students' smartphones.

The system ensures secure authentication, role-based access (Admin & Student), and real-time updates using Firebase services.

Why College News Hub App?

1. **Instant Notifications** – Important notices reach students immediately.
2. **Paperless System** – Reduces physical notice board dependency.
3. **Centralized Information** – All updates in one place.
4. **Secure Access** – Admin-controlled posting system.
5. **Organized Categories** – Easy search and filter options.

2. Literature Survey

Sr. No.	Title of Paper	Author Name	Findings	Remark
1	Mobile-Based Notice Systems	A. Sharma	Digital notice boards improve communication efficiency.	Relevant
2	Cloud-Based Campus Applications	R. Kulkarni	Firebase provides scalable backend for apps.	Useful
3	Push Notification Systems in Education	P. Verma	FCM improves engagement and response rate.	Applicable

3. Proposed work

The proposed system, **College News Hub App**, is a complete Android application that allows colleges to digitally manage and broadcast information.

The system consists of:

User Roles

1. Admin

- Add news
- Edit news
- Delete news
- Upload images/PDF links
- Send announcements
- Mark important notices

2. Student/User

- View all news
- Read detailed notices
- Search by title/category/department
- Filter updates
- Bookmark notices
- Receive push notifications

Key Features

• Secure Login & Registration using Firebase Authentication
• Real-time Firestore database
• Firebase Storage for image & PDF
• Push notifications using FCM
• RecyclerView based dynamic feed
• Dark Mode support
• Pull-to-refresh functionality
• Share & Bookmark options

4. Architecture and working

System Architecture

1. **User Interface Layer**
    a. XML Layouts
    b. Material UI Components
    c. Activities & RecyclerView
2. **Application Layer**
    a. Java Classes
    b. OOP Implementation
    c. Role Management
    d. Business Logic
3. **Firebase Backend**
    a. Firebase Authentication
    b. Firebase Firestore Database
    c. Firebase Storage
    d. Firebase Cloud Messaging
4. **Notification Service**
    a. FCM token generation
    b. Topic-based notifications
    c. Admin-triggered alerts

Working Steps

1. User installs and opens the application.
2. Splash Screen loads.
3. User registers or logs in.
4. Firebase authenticates user.
5. Role is verified (Admin/Student).
6. Home feed loads news from Firestore.
7. RecyclerView displays news items.
8. Clicking item opens detailed activity.
9. Admin can create/edit/delete posts.
10. Notifications are sent using FCM.

1. Applications, Advantages, Disadvantages

Applications

- Educational Institutions
- Colleges and Universities
- Coaching Centers
- Training Institutes

Advantages

- Real-time updates
- Secure login system
- Organized information
- Paperless communication
- Push notifications
- Easy search and filter

Disadvantages

- Requires internet connection
- Firebase dependency
- Admin misuse possibility (if not controlled)

2. Future Scope

- Integration with Attendance System
- Voice-based notice reading
- AI-based personalized notification system
- Multi-language support
- Offline caching system
- Web dashboard for Admin

3. Conclusion

The **College News Hub App** provides a modern, secure, and efficient digital solution for managing college announcements.

Using Java and Firebase, the application ensures real-time communication, secure authentication, structured data storage, and push notification delivery.

The system improves communication transparency between administration and students, reduces paper usage, and ensures important updates are never missed.

With future enhancements, the system can evolve into a complete campus management platform.

References

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